

Algebraic Geometry

Exercise 2

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2.1 Problem

Let R be a ring, and set $X = \text{Spec}(R)$.

- (1) Let $I, J \leq R$ be ideals. Show that if $I \subseteq J$ then $V(J) \subseteq V(I)$, and give an example where the converse fails.
- (2) Let $Z, W \subseteq X$ be subsets. Show that if $Z \subseteq W$ then $I(W) \subseteq I(Z)$, and give an example where the converse fails.
- (3) Show that $V(I(Z)) = \overline{Z}$ for all $Z \subseteq X$, and $I(V(J)) = \sqrt{J}$ for all ideals $J \leq R$.

- (1) $V(J)$ is the set of all prime ideals containing J . Since I is contained in J , a prime ideal which contains J must contain I , and thus is in $V(I)$. In the case of $R = \mathbb{Z}$, let $I = 2\mathbb{Z}$ and $J = 4\mathbb{Z}$ so that $V(I) = V(J) = \{2\mathbb{Z}\}$, but clearly we have that I is not a subset of J .
- (2) $I(W)$ is the set of all $f \in R$ such that $W \subseteq V(f)$. Since Z is contained in W , this implies $Z \subseteq V(f)$ and so $f \in I(Z)$, as desired. Take $R = \mathbb{C}[x]$, whose spectrum consists of the zero ideal and all principal ideals of the form $(x - \lambda)$ for $\lambda \in \mathbb{C}$. Let $W = \{0\}$ contain just the zero ideal and $Z = \{(x - \lambda)\}_{\lambda \in \mathbb{C}}$ be its complement. Then clearly we don't have $Z \subseteq W$, but

$$I(W) = \bigcap W = 0, \quad I(Z) = \bigcap_{\lambda \in \mathbb{C}} Z = \bigcap_{\lambda \in \mathbb{C}} (x - \lambda) = 0,$$

since a polynomial is in $(x - \lambda)$ iff λ is a root of it, and if it is in all $(x - \lambda)$ then it must be zero. Thus $I(W) \subseteq I(Z)$, Z is not a subset of W .

- (3) The closed sets of X are the vanishing sets $V(\bullet)$, so the closure of Z is of this form. $V(I(Z))$ is the set of all prime ideals containing $I(Z) = \bigcap_{\mathfrak{p} \in Z} \mathfrak{p}$. If $Z \subseteq V(J)$ then for every $\mathfrak{p} \in Z$, we have that $J \leq \mathfrak{p}$, which means that $J \leq \bigcap_{\mathfrak{p} \in Z} \mathfrak{p} = I(Z)$. We showed above that $V(\bullet)$ is contravariant, so $V(I(Z)) \subseteq V(J)$, meaning $V(I(Z))$ is the smallest closed set containing Z ; it is its closure.

By definition,

$$I(V(J)) = \bigcap_{\mathfrak{p} \in V(J)} \mathfrak{p} = \bigcap_{J \leq \mathfrak{p}} \mathfrak{p} = \sqrt{J},$$

as desired.

2.2 Problem

Let R, S be rings, $X = \text{Spec}(R), Y = \text{Spec}(S)$ their spectra, and $\phi: R \rightarrow S$ a ring morphism with $\phi^*: Y \rightarrow X$ the induced map.

- (1) Prove that $\ker \phi \subseteq \text{nil}(R)$ iff $\phi^*(Y)$ is dense in X . Find an example where ϕ is injective, but $\phi^*(Y) \neq X$.
- (2) Assume that ϕ is surjective. Prove that ϕ^* is a closed immersion (its image is closed, and it is a homeomorphism onto its image).

- (1) $\phi^*(Y)$ is dense in X iff its closure is X . We showed that the closure of a subset of X is equal to $V(I(\bullet))$, thus $\phi^*(Y)$ is dense iff $V(I(\phi^*Y)) = X$. Since $X = V(0)$, and the vanishing sets of two

ideals are the same iff their radicals are, this occurs iff $\sqrt{I(\phi^*Y)} = \text{nil}(R)$. Now $\sqrt{I} = \text{nil}(0)$ iff $I \leq \text{nil}(0)$, since the radical is order-preserving and $I \leq \sqrt{I}$. So ϕ^*Y is dense iff $I(\phi^*Y) \leq \text{nil}(R)$.

Now ϕ^*Y is the set of all preimages of prime ideals in S , so

$$I(\phi^*Y) = \bigcap_{\mathfrak{q} \in \text{Spec}(S)} \phi^{-1}\mathfrak{q} = \phi^{-1} \bigcap_{\mathfrak{q} \in \text{Spec}(S)} \mathfrak{q} = \phi^{-1}\text{nil}(S).$$

So ϕ^*Y is dense iff $\phi^{-1}\text{nil}(S) \leq \text{nil}(R)$. If $\ker\phi \subseteq \text{nil}(R)$ then if $f \in \phi^{-1}\text{nil}(S)$ we have that $\phi(f^n) = 0$ for some n , which means $f^n \in \ker\phi \subseteq \text{nil}(R)$ so that $f \in \text{nil}(R)$; meaning $\phi^{-1}\text{nil}(S) \leq \text{nil}(R)$ as desired. Conversely let $\phi(f) = 0$ then $f \in \phi^{-1}\text{nil}(S) \leq \text{nil}(R)$ so $\ker\phi \leq \text{nil}(R)$ as desired.

Let k be a field of zero characteristic and $\phi: \mathbb{Z} \rightarrow k$ the unique morphism, which is injective by virtue of k having zero characteristic. We have that $X = \text{Spec}(\mathbb{Z})$ is infinite, yet $Y = \text{Spec}(k)$ just consists of a single point. Thus $\phi^*: Y \rightarrow X$ cannot be surjective.

- (2) If ϕ is surjective, then it induces an isomorphism $R/\ker\phi \rightarrow S$, which maps prime ideals of $R/\ker\phi$ to S . Prime ideals of $R/\ker\phi$ are in bijection with prime ideals of R containing $\ker\phi$, i.e. $V(\ker\phi)$. The inverse of this, the map $Y \rightarrow V(\ker\phi)$ is given by ϕ^{-1} . And so we see that $\phi^*Y = V(\ker\phi)$, so ϕ^*Y is closed.

To show that ϕ^* is a homeomorphism onto its image, it is sufficient to show that it is injective and a closed map onto its image. Indeed, for an ideal $J \leq S$,

$$\phi^*V(J) = \{\phi^{-1}\mathfrak{q} \mid J \leq \mathfrak{q}\},$$

and by surjectivity, $J \leq \mathfrak{q}$ iff $\phi^{-1}J \leq \phi^{-1}\mathfrak{q}$, so this is equal to

$$\phi^*V(J) = \{\phi^{-1}\mathfrak{q} \mid \phi^{-1}J \leq \phi^{-1}\mathfrak{q}\} = V(\phi^{-1}J) \cap \phi^*Y,$$

which is closed in its image (since $\phi^*Y = V(\ker\phi)$, this is equal to $V(\phi^{-1}J)$, but that's not important here.) Injectivity follows from surjectivity of ϕ : the preimage map of a surjective function is injective.

2.3 Problem

Let $S \subseteq R$ be multiplicatively closed, and denote by $\iota: R \rightarrow S^{-1}R$ the structure map.

- (1) Show that ι^* is a topological embedding.
- (2) S is **saturated** if $xy \in S$ implies $x \in S$ and $y \in S$.
 - (i) S has a **saturation**: a smallest saturated multiplicatively closed set containing it.
 - (ii) Let $I \leq R$ be an ideal. Show that $S = 1 + I$ is multiplicatively closed, and compute its saturation.
 - (iii) Show that if $\hat{S}$ and S have the same saturation, then $S^{-1}R \cong \hat{S}^{-1}R$.
 - (iv) Show that S is saturated iff $R - S = \bigcup_{S \cap \mathfrak{p} = \emptyset} \mathfrak{p}$, where the intersection runs over prime ideals of R which don't intersect S .
 - (v) Show that the set of zero-divisors of R is a union of prime ideals.

- (1) ι^* is already continuous. In lecture we showed that $\mathfrak{q} \mapsto \iota^*\mathfrak{q}$ is a bijection from $\text{Spec}(S^{-1}R)$ to prime ideals $\mathfrak{p} \leq R$ not intersecting S . So ι^* is injective, and to show that it is a homeomorphism onto its image it is sufficient to show it is a closed map. We know that ι^* 's inverse is S^{-1} , and it extends to a left inverse on all ideals of $S^{-1}R$: $S^{-1}\iota^{-1}J = J$ for all ideals $J \leq S^{-1}R$, in particular all ideals of $S^{-1}R$ are of the form $S^{-1}I$ for an ideal $I \leq R$.

To show ι^* is closed, we compute its image on $V(S^{-1}I)$:

$$\iota^*V(S^{-1}I) = \{\iota^{-1}\mathfrak{q} \mid S^{-1}I \leq \mathfrak{q}\}.$$

Now we claim that $S^{-1}I \leq \mathfrak{q}$ iff $I \leq \iota^*\mathfrak{q}$. If $S^{-1}I \leq \mathfrak{q}$ and $x \in I$, then $\iota(x) \in S^{-1}I \leq \mathfrak{q}$ and so $x \in \iota^*\mathfrak{q}$. Conversely, since S^{-1} is order-preserving, if $I \leq \iota^*\mathfrak{q}$ then $S^{-1}I \leq S^{-1}\iota^*\mathfrak{q} = \mathfrak{q}$ as desired. Thus

$$\iota^*V(S^{-1}I) = \{\iota^*\mathfrak{q} \mid I \leq \iota^*\mathfrak{q}\} = V(I) \cap \text{im } \iota^*,$$

which is closed in the subspace topology. Thus ι^* is indeed a topological embedding.

(2)

- (i) We can give an easy nonconstructive proof by noting that the arbitrary intersection of saturated multiplicatively closed sets is saturated and multiplicatively closed. Since $S \subseteq R$ is saturated and multiplicatively closed, taking the intersection of all saturated multiplicatively closed sets containing S will yield the desired set.

Constructively, we claim that

$$\hat{S} = \{f \in R \mid fg \in S \text{ for some } g \in R\}$$

is the desired saturation. Firstly, it is multiplicatively closed: for $f, g \in \hat{S}$ suppose that f', g' satisfy $ff', gg' \in S$. Then $fgf'g' \in S$ so that $fg \in \hat{S}$. And it is saturated: if $fg \in \hat{S}$ then there is a $h \in R$ such that $fgh \in S$. Since $fgh = f(gh) = g(fh)$, we see that by definition $f, g \in \hat{S}$ as desired. Clearly we also have that $S \subseteq \hat{S}$, as for $f \in S$, $f \cdot 1 \in S$.

Now if $S \subseteq S'$ is saturated, then for $f \in \hat{S}$ we have that $fg \in \hat{S}$ for some $g \in R$. Then $fg \in S'$ and so $f \in S'$, meaning $\hat{S} \subseteq S'$ as desired. Thus $\hat{S}$ is the desired saturation.

- (ii) S is multiplicatively closed: for $x, y \in I$ we have that $(1+x)(1+y) = 1+x+y+xy$ which is in $1+I$. The above point shows us that S 's saturation is

$$\hat{S} = \{f \in R \mid fg \in 1+I \text{ for some } g \in R\},$$

i.e. the set of all $f \in R$ which are invertible in R/I . So $\hat{S} = \pi^{-1}(R/I)^\times$, where $\pi: R \rightarrow R/I$ is the projection.

Now in general, $x \in R$ is invertible iff it is not contained in any proper ideal. That is,

$$R^\times = \bigcap_{I < R} I^c.$$

This intersection is equal to the union over all prime ideals: clearly the intersection over all prime ideals is a superset of this, and conversely every proper ideal has an extension to a prime ideal, so the intersections are the same.

The prime ideals of R/I are in bijection with the prime ideals containing I , $V(I)$. Thus

$$\hat{S} = \pi^{-1} \bigcap_{\mathfrak{p} \in V(I)} (\mathfrak{p}/I)^c = \bigcap_{\mathfrak{p} \in V(I)} \mathfrak{p}^c.$$

We can further consider only maximal ideals: since every ideal has an extension to a maximal ideal, $R^\times$ is the intersection over all maximal extensions of I . So we also have

$$\hat{S} = \bigcap_{\mathfrak{m} \in V_m(I)} \mathfrak{m}^c,$$

where $V_m(I) = V(I) \cap \text{Specm}(R)$ is the vanishing set of maximal ideals.

- (iii) It is sufficient to prove this when $\hat{S}$ is S 's saturation. Consider the structure morphism $\iota: R \rightarrow S^{-1}R$. Notice then that we have the following: if $\iota(r) = 0$ then there is a $s \in S \subseteq \hat{S}$ such that $sr = 0$, and every element of $S^{-1}R$ is of the form $\iota(r)\iota(s)^{-1}$ for some $r \in R, s \in S \subseteq \hat{S}$. Thus we need only show that $\iota(\hat{S})$ are invertible to apply a result from lecture giving us an isomorphism $\hat{S}^{-1}R \cong S^{-1}R$.
- For $f \in \hat{S}$, there is a $g \in R$ such that $fg \in S$, which means that $\iota(fg) = \iota(f)\iota(g)$ is invertible. This implies that $\iota(f)$ is invertible as well (its inverse is $\iota(g)\iota(fg)^{-1}$), as desired.
- (iv) Notice that for a prime $\mathfrak{p}$, $\mathfrak{p}^c$ is saturated: if $fg \notin \mathfrak{p}$ then $f, g \notin \mathfrak{p}$ since it is an ideal. And as noted above, the intersection of saturated sets is saturated. Thus if $R - S$ is a union of prime ideals, then S is an intersection of the complement of prime ideals, so the intersection of saturated sets, and is thus saturated.
- Notice that the saturation of S is equal to the preimage under ι of the units of $S^{-1}R$. Indeed, if $fg \in S$ then $\iota(fg) \subseteq \iota(S)$ is a unit, so $\iota(f)$ is. Conversely if $\iota(f)$ is a unit then there is a $g \in R, s, t \in S$ such that $t(fg - s) = 0$, which implies $tfg \in S$, so that f is in the saturation.
- Thus if S is saturated and $f \in R - S$ then $\iota(f)$ is not a unit, and thus is contained in a prime ideal $\mathfrak{p} \leq S^{-1}R$. Then $f \in \iota^*\mathfrak{p}$ is a prime ideal not intersecting S , so f is in the union. And if $f \in \mathfrak{p}$ for $\mathfrak{p} \cap S = \emptyset$ then $f \notin S$ so the inclusion in the other direction is true always.
- (v) Let S be the set of non-zero-divisors of R . This is a multiplicatively closed saturated set: if $f, g \in S$ are non-zero-divisors so is fg since if $fgh = 0$ then $gh = 0$ and so $h = 0$; and if $fg \in S$, then $f, g \in S$ since if $fh = 0$ then $fgh = 0$ so $h = 0$. Thus $R - S$, the set of zero-divisors, is a union of prime ideals, as above.

2.4 Problem

Let R be a ring and $X = \text{Spec}(R)$.

- (1) Let $\mathfrak{p} \in X$, show that $R_{\mathfrak{p}}$ is a local ring, and its maximal ideal is $\mathfrak{p}R_{\mathfrak{p}}$.
- (2) Consider the case $R = \mathbb{C}[x]$ and $\mathfrak{p} = (x)$. Describe the topological space $\text{Spec}(R_{\mathfrak{p}})$, and the map $\text{Spec}(R_{\mathfrak{p}}) \rightarrow X$ corresponding to the structure map, explicitly.
- (3) Let $\mathfrak{p} \in X$, define the **residue field** of $\mathfrak{p}$ on X to be $k(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$. Prove that this is indeed a field.
- (4) Let k be any field. Show that a morphism $\text{Spec}(K) \rightarrow X$ (i.e. a morphism $R \rightarrow K$) is the same as giving a point $\mathfrak{p} \in X$ and an embedding $k(\mathfrak{p}) \rightarrow K$.

- (1) We show that $\mathfrak{p}R_{\mathfrak{p}}$'s complement consists of units. Indeed, $\mathfrak{p}R_{\mathfrak{p}}$ consists of all fractions x/y with $x \in \mathfrak{p}$ and $y \notin \mathfrak{p}$. So an element of its complement is of the form x/y with $x, y \notin \mathfrak{p}$ which is a unit as it has inverse y/x . And $\mathfrak{p}R_{\mathfrak{p}}$ is proper since $\mathfrak{p}$ intersects with the localizing set trivially (the set is $\mathfrak{p}$'s complement). Thus $\mathfrak{p}R_{\mathfrak{p}}$ is the unique maximal ideal of $R_{\mathfrak{p}}$, as desired.
- (2) Prime ideals in $\text{Spec}(R_{\mathfrak{p}})$ are of the form $(R - \mathfrak{p})^{-1}\mathfrak{q} = \mathfrak{q}R_{\mathfrak{p}}$ for prime $\mathfrak{q}$ where $\mathfrak{q} \cap (R - \mathfrak{p}) = \emptyset$, i.e. $\mathfrak{q} \subseteq \mathfrak{p}$. That is,

$$\text{Spec}(R_{\mathfrak{p}}) = \{\mathfrak{q}R_{\mathfrak{p}} \mid \mathfrak{q} \subseteq \mathfrak{p}\}.$$

In our case, $\mathfrak{p} = (x)$ only contains two prime ideals: itself and zero. Thus

$$Y = \text{Spec}(\mathbb{C}[x]_{(x)}) = \{0, (x)\mathbb{C}[x]_{(x)}\}.$$

The closed sets of this space are $\emptyset, Y, \{(x)\mathbb{C}[x]_{(x)}\}$. The singleton containing the maximal ideal is the vanishing set of the maximal ideal, and $\{0\}$ is not a vanishing set since if $I \leq 0$ then $I = 0$ and so $V(I) = Y$. Thus the open sets are $\emptyset, Y, \{0\}$.

The morphism $\iota^*: Y \rightarrow X$ corresponding to the structure map maps 0 to $\iota^{-1}0$ which is the zero ideal in $\mathbb{C}[x]$. And it maps $(x)\mathbb{C}[x]_{(x)}$ to $(x) \in \mathbb{C}[x]$ (since $\iota^{-1}S^{-1}\mathfrak{p} = \mathfrak{p}$ for prime $\mathfrak{p}$).

- (3) This is just the quotient of a ring by a maximal ideal; this is always a field.
- (4) Let $\phi: R \rightarrow K$ be a morphism, then since $\text{Spec}(K) = \{0\}$, the induced morphism $\phi^*: \text{Spec}(K) \rightarrow \text{Spec}(R)$ corresponds just to the point $\phi^{-1}0 = \ker \phi = \mathfrak{p}$. To then define an inclusion $k(\mathfrak{p}) \rightarrow K$, we need to find a map $R_{\mathfrak{p}} \rightarrow K$ where $\mathfrak{p}R_{\mathfrak{p}}$ is contained in the kernel. For a map $R_{\mathfrak{p}} \rightarrow K$, we need a map $R \rightarrow K$ which maps $R - \mathfrak{p}$ to units, in our case just non-zero values. By definition ϕ does this, and thus it defines a map $\tilde{\phi}: R_{\mathfrak{p}} \rightarrow K$ by $x/y \mapsto \phi(x)\phi(y)^{-1}$. Its kernel consists of fractions of the form x/y for $x \in \ker \phi = \mathfrak{p}$, so $\mathfrak{p}R_{\mathfrak{p}}$ is indeed contained in the kernel. Thus we can quotient $\tilde{\phi}: R_{\mathfrak{p}} \rightarrow K$ to give a ring morphism $k(\mathfrak{p}) \rightarrow K$, which is immediately an embedding as it originates from a field. Explicitly, $\tilde{\phi}$ is defined by

$$\tilde{\phi}(r/x + \mathfrak{p}R_{\mathfrak{p}}) = \phi(r)\phi(x)^{-1}.$$

Now, let $\psi: k(\mathfrak{p}) \rightarrow K$ be an embedding. We define $\hat{\psi}: R \rightarrow K$ via the following composition:

$$\hat{\psi}: R \xrightarrow{\iota} R_{\mathfrak{p}} \xrightarrow{\pi} k(\mathfrak{p}) \xrightarrow{\psi} K.$$

Explicitly, $\hat{\psi}(r) = \psi(r/1 + \mathfrak{p}R_{\mathfrak{p}})$.

We claim that the two maps defined above: mapping ϕ to $\tilde{\phi}$ and vice versa, are inverses. Indeed, if we are given ϕ and we define $\tilde{\phi}$, then

$$\hat{\tilde{\phi}}(r) = \tilde{\phi}(r/1 + \mathfrak{p}R_{\mathfrak{p}}) = \phi(r)\phi(1)^{-1} = \phi(r).$$

And if we are given ψ and we define $\hat{\psi}$, then $\ker \hat{\psi}$ are all $r \in R$ such that $r/1 + \mathfrak{p}R_{\mathfrak{p}} \in \ker \psi$. Since ψ is an embedding, its kernel is trivial and thus $r/1 \in \mathfrak{p}R_{\mathfrak{p}}$, meaning $r \in \mathfrak{p}$ (by saturation). Thus $\ker \hat{\psi} = \mathfrak{p}$, and

$$\tilde{\hat{\psi}}[r/x] = \hat{\psi}(r)\hat{\psi}(x)^{-1} = \psi[r/1] \cdot \psi[x/1]^{-1},$$

where $[\bullet] = \bullet + \mathfrak{p}R_{\mathfrak{p}}$ are the cosets. Since $x/1$ is invertible, so is $[x/1]$ with inverse $[1/x]$, so this is equal to

$$\tilde{\hat{\psi}}[r/x] = \psi[r/1]\psi[1/x] = \psi[r/x].$$

Thus $\tilde{\bullet}$ and $\hat{\bullet}$ are the desired bijections.

2.5 Problem

Let $f \in \mathbb{C}[x, y]$ be a nonzero irreducible polynomial with $f(0, 0) = 0$. Define $R = \mathbb{C}[x, y]/(f)$ and $X = \text{Spec}(R)$; X is called a **plane curve**. X is **singular at the origin** if $\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0) = 0$.

- (1) Let $\iota: R \rightarrow R_x$ be the structure map. Prove that if $y/x \in \text{im } \iota$ then X is nonsingular at the origin.
- (2) The converse is not true. There are two problems:
 - (i) The tangent line to X at the origin may be vertical. For example, let $f = x - y^2$. Show that X is nonsingular at the origin, but $y/x \notin \text{im } \iota$.

- (ii) y/x may have poles other than at the origin. For example, let $f = y^2 - y + x^2$. Show that X is nonsingular at the origin, but $y/x \notin \text{im}\iota$.
- (3) Suppose $\frac{\partial f}{\partial y}(0,0) \neq 0$, and let $\iota': R_{(x,y)} \rightarrow \text{Frac}(R)$ be the structure morphism. Show that $y/x \in \text{im}\iota'$.

- (1) If $y/x \in \text{im}\iota$ then there is a $g \in R$ such that $y/x = g/1$. Since R is an integral domain, as f is irreducible, this means that $y = xg$ modulo (f) . Writing $g = g + (f)$, this means that $xg - y \in (f)$, so there is a polynomial h such that $xg - y = fh$. Applying the product rule for gradients, we see that

$$\nabla(xg - y)(0) = \nabla(xg)(0) - \nabla(y)(0) = g(0)e_1 - e_2,$$

where e_i are the standard basis vectors. On the other hand,

$$\nabla(fh)(0) = f(0)\nabla h(0) + h(0)\nabla f(0) = h(0)\nabla f(0).$$

Thus we have that $h(0)\nabla f(0) = g(0)e_1 - e_2$ which is nonzero, so $\nabla f(0)$ is nonzero, meaning X is nonsingular at the origin.

(2)

- (i) f 's partial derivative with respect to x is 1, so X is nonsingular at the origin. But as above, if $y/x \in \text{im}\iota$ we have that $xg - y = fh$ for some polynomials g, h . But then we have that, setting $x = 0$, $-y = h(0, y) \cdot (-y^2)$. Considering dimensions, this cannot be.
- (ii) f 's partial derivative with respect to y is $2y - 1$, which is -1 at the origin, so X is nonsingular at the origin. And again, if $y/x \in \text{im}\iota$ then $xg - y = fh$ for some polynomials g, h . But then setting $x = 0$ and $y = 1$ gives us $-1 = 0$, a contradiction.
- (3) We want to show that $y/x = g/h$ modulo f for some $h \notin (x, y)$, that is $hy - gx \in (f)$. Suppose f is of the form

$$f = \sum_{t,\ell} a_{t,\ell} x^t y^\ell.$$

Since $f(0,0) = 0$ we can take the sum over t, ℓ not both zero. Then let us write

$$f = y \sum_{\ell \geq 1, t} a_{t,\ell} x^t y^{\ell-1} + x \sum_{t \geq 1} a_{t,0} x^{t-1}.$$

f 's partial derivative with respect to y at 0 is

$$\frac{\partial f}{\partial y}(0,0) = \sum_{t,\ell} a_{t,\ell} \ell x^t y^{\ell-1}(0,0) = a_{0,1}.$$

So let $h = \sum_{\ell \geq 1, t} a_{t,\ell} x^t y^{\ell-1}$ and $g = -\sum_{t \geq 1} a_{t,0} x^{t-1}$, so that $f = hy - gx$. And notice that $h(0,0) = a_{0,1} \neq 0$ so $h \notin (x, y)$, as desired.

2.6 Problem

Let $X = \text{Spec}(R)$ be the spectrum of a ring. Then V forms a contravariant functor $\mathcal{P}(R) \rightarrow \mathcal{P}(X)$,

and I forms a contravariant functor $\mathcal{P}(X) \rightarrow \mathcal{P}(R)$. Show that these form an antitone Galois connection: viewed as functors $V: \mathcal{P}(R) \rightleftarrows \mathcal{P}(X)^{\text{op}}: I$, V is a left adjoint of I .

Recall that a functor between posets is just an order-preserving function. If P, Q are posets, an adjunction $F: P \rightleftarrows Q: G$ amounts to two functors satisfying either of the two equivalent conditions:

- (1) $Fp \leq q$ iff $p \leq Gq$ for all $p \in P$ and $q \in Q$. This is the Yoneda form of an adjunction; a natural bijection between hom-sets.
- (2) $p \leq GFp$ and $FGq \leq q$ for all $p \in P$ and $q \in Q$. This is the unit-counit form of an adjunction; a natural map from a poset functor to another amounts to a statement that the first is less than the other, and the triangle identities always hold in a poset category.

In our case, these amount to: $Z \subseteq V(J)$ iff $J \subseteq I(Z)$ (by contravariance), and $J \subseteq I(V(J))$ and $Z \subseteq V(I(Z))$. The latter is easy to see, as we showed above that $I(V(J)) = \sqrt{J}$ and $V(I(Z)) = \overline{Z}$. The former is also easy to see: if $Z \subseteq V(J)$ then for $f \in J$ we have that $Z \subseteq V(J) \subseteq V(f)$ so that $f \in I(Z)$. Conversely if $J \subseteq I(Z) = \bigcap_{\mathfrak{p} \in Z} \mathfrak{p}$, then $J \subseteq \mathfrak{p}$ for all $\mathfrak{p} \in Z$ so $\mathfrak{p} \in V(J)$, so $Z \subseteq V(J)$ as desired.

Thus we have shown, in two ways, that V, I forms an antitone Galois connection.

2.7 Problem

Let X be a compact Hausdorff space, and define $R = C(X)$ to be the ring of continuous real-valued functions on X . Further define $Y = \text{Specm}(R)$ to be the maximum spectrum of R .

- (1) For $x \in X$, show that $I(x) = \{f \in R \mid f(x) = 0\}$ is a maximal ideal of R . This defines a map $\mu: X \rightarrow Y$.
- (2) Show that μ is continuous.
- (3) Let $\mathfrak{m} \in Y$, and define $V(\mathfrak{m}) = \{x \in X \mid f(x) = 0 \text{ for all } f \in \mathfrak{m}\}$.
 - (i) Show that $V(\mathfrak{m}) \neq \emptyset$.
 - (ii) Conclude that μ is surjective.
- (4) Use Urysohn's lemma to show that μ is injective.
- (5) For $f \in R$, define $X_f = \{x \in X \mid f(x) \neq 0\} = R - V(x)$. Show that X_f forms a basis for the topology of X , and $\mu(X_f) = Y_f$.
- (6) Conclude that μ is a homeomorphism.

- (1) If we consider the map $\text{ev}_x: R \rightarrow \mathbb{R}$ mapping a function f to its value on x , we see that $I(x)$ is simply the kernel of ev_x , and is thus an ideal. Now, if $c = g(x) \neq 0$ then consider the function $f = c - g$, which is in $V(x)$, and we see that $f + g = c$ is invertible (with inverse c^{-1}). Thus $(g) + I(x)$ contains a unit, and thus is all of R . So $I(x)$ must be maximal.
- (2) Let $V(J) \in Y$ be closed, then its preimage under μ is

$$\mu^{-1}V(J) = \{x \in X \mid J \leq I(x)\},$$

which we claim is closed. Indeed, $J \leq I(x)$ iff for every $f \in J$, $f(x) = 0$, iff $x \in f^{-1}0$. So this is equal to

$$\mu^{-1}V(J) = \bigcap_{f \in J} f^{-1}0,$$

an intersection of closed sets, and is thus closed as desired. So the preimage under μ of closed sets is closed, so μ is continuous as desired.

(3)

(i) Notice that

$$V(\mathfrak{m}) = \bigcap_{f \in \mathfrak{m}} \{x \in X \mid f(x) = 0\} = \bigcap_{f \in \mathfrak{m}} f^{-1}0.$$

Since X is compact, if $V(\mathfrak{m}) = \emptyset$ then there must be a finite collection of elements $f_1, \dots, f_n$ such that $\bigcap_{i=1}^n f_i^{-1}0 = \emptyset$. Then the function $g = f_1^2 + \dots + f_n^2$ is continuous and has no zeroes (a zero of g must be a zero of every f_i). Since inversion is continuous $\mathbb{R}^\times \rightarrow \mathbb{R}^\times$, g has an inverse. But $g \in \mathfrak{m}$, a contradiction to $\mathfrak{m}$ being a proper ideal.

(ii) As above, $V(\mathfrak{m}) \neq \emptyset$, so let $x \in V(\mathfrak{m})$. Then $\mathfrak{m} \subseteq \mu(x)$, and by maximality this means $\mu(x) = \mathfrak{m}$. So μ is surjective.

- (4) For $x \neq y \in X$, since X is compact Hausdorff and thus normal, there must be a Urysohn function $f: X \rightarrow \mathbb{R}$ such that $f(x) = 0, f(y) = 1$. But then $f \in \mu(x)$ and $f \notin \mu(y)$ so $\mu(x) \neq \mu(y)$ and μ is injective.
- (5) It is sufficient to show that $V(f) = f^{-1}0$ forms a basis of closed sets. Clearly $V(f)$ are all closed as the preimage of closed sets. Now, $\emptyset = V(1)$ and $X = V(0)$ so we need only show that all nontrivial proper closed subsets of X are intersections of sets of the form $V(f)$. For $S \subsetneq X$ and for each $x \notin S$, let $f_x \in R$ be a Urysohn function where $f_x(S) = 0$ and $f_x(x) = 1$. Then we claim

$$S = \bigcap_{x \notin S} V(f_x).$$

Indeed, for each $x \notin S$, $S \subseteq f_x^{-1}0 = V(f_x)$ so that S is a subset of the intersection. And for $x \notin S$, $x \notin V(f_x)$ and so x is not in the intersection. Thus S can be written as an intersection of closed basis sets.

Since μ is a bijection, we show instead that $\mu(V(f)) = V_Y(f) = \{\mathfrak{m} \in Y \mid f \in \mathfrak{m}\}$. Indeed, since μ is a bijection every maximal ideal in Y is of the form $\mu(x)$, so

$$V_Y(f) = \{\mathfrak{m} \in Y \mid f \in \mathfrak{m}\} = \{\mu(x) \mid f \in \mu(x)\} = \{\mu(x) \mid f(x) = 0\} = \mu(V(f)),$$

as desired. As μ is a bijection it preserves complements so $\mu(X_f) = \mu(V(f)^c) = V_Y(f)^c = Y_f$ as desired.

- (6) Since μ maps open basis sets to open sets in Y , it maps open sets to open sets and is thus an open map. It is also a continuous bijection, thus a homeomorphism.